



TFRA: Trajectory-Based Message Ferry Recognition Attack in UAV Network

Yuting Wu¹(✉), Yulei Liu¹, Liang Liu¹(✉), Feng Wang¹, Lihong Fan¹,
and Qian Zhou²

¹ Nanjing University of Aeronautics and Astronautics, Nanjing, China
{yutingwu, liu_yulei, liangliu, lih}@nuaa.edu.cn

² Nanjing University of Posts and Telecommunications, Nanjing, China
zhouqian@njupt.edu.cn

Abstract. The introduction of ferry UAVs (message ferry) in the UAV network is an effective means to solve the cooperative communication of multiple UAVs. The ferry UAVs act as message collectors and throwers in the network to improve data transmission efficiency, the critical roles that ferry plays have made them the target of sophisticated attacks, ferry should be protected, otherwise, routing efficiency and data delivery of the network will be affected. Since the movement mode of ferry nodes is different from the ordinary nodes, this paper attempts to distinguish ordinary node and ferry node through trajectory analysis, then we propose a Trajectory-based message Ferry Recognition Attack (TFRA) which is easy for an attacker to implement, using the idea of trajectory clustering to distinguish the trajectory of the ferry node, and obtain the location for further attacks. At the same time, we conducted a systematic study on the existing path planning scheme of ferry node, and summarized several typical types of message ferry path planning schemes, then evaluated the performance of TFRA in these schemes. The results show TFRA can attack ferry nodes with high accuracy and recognition rate.

Keywords: UAV network · Message ferry recognition · Trajectory clustering

1 Introduction

In recent years, UAV (unmanned aerial vehicle) has become a promising new computing platform and has broad application prospects on search and rescue operations [18], forest fire automatic monitoring [20], ground target detection and tracking [29], and disaster monitoring [19], etc. The cooperative application of small multi-UAVs has the advantages of greater survivability, higher scalability, faster completion of tasks, smaller radar cross-sections, but it also faces some

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unique challenges. Among them, one of the most important design issues is collaborative communication between multiple UAVs.

It is an attractive scheme to establish connections by introducing message ferry. Message ferry is a mobile assistive technology whose existence promotes connections and solves the disconnection and sparsity of mobile ad hoc networks [11]. Message ferry is a delay-tolerant forwarding scheme that physically transfers data to the destination or the next relay node without an end-to-end routing path [21]. By using the ferry node as a relay, the node can communicate asynchronously with other disconnected nodes, or transmit the collected data to the ground station when other nodes perform normal data collection, which can increase routing path diversity and network delivery rate [7].

In a UAV network containing ferry node, the ferry node acts as data collector and data transmitter, its movement mode is well-designed to meet the needs of the network, and the critical roles that ferry nodes play have made them the target of sophisticated attacks, therefore its location anonymity is very important. If attacker attacks the ferry node, making it unable to work, then the routing efficiency of the network reduces. Or the ferry node is captured, attackers can carry out attacks such as the dissemination of false information, black hole, and gray hole attacks [4], making the network paralyzed or unable to work.

Using technology such as radar, an attacker can easily obtain UAV's position and depict their flight trajectory [10]. In the actual application of UAVs, the network may include different types of UAVs, such as ferry UAVs and ordinary UAVs. In this case, if the attacker can also distinguish the types of nodes in the network, they can easily attack important nodes such as ferry nodes to achieve the purpose of paralyzing the UAV network at a lower cost.

There have been many studies on the location anonymity of nodes in wireless sensor networks, and many kinds of attack methods and location anonymity protection methods have also been proposed. As far as we know, the research on ferry nodes' location anonymity in UAV network is still an open problem. In this paper, we aim to identify and attack ferry nodes in UAV network, our contributions can be summarized as follows:

- We systematically studied path planning methods of ferry nodes in UAVs and self-organizing networks, and have summarized several typical path planning methods of ferry nodes according to their main design ideas.
- The flight mode of the ferry node is different from the ordinary node. Based on this feature, we propose a Trajectory-based message Ferry Recognition Attack (TFRA). We design a global attacker who can observe near the UAV network and obtain the flight trajectory of UAVs through radar and other technologies, then the attacker identifies ferry nodes through trajectory clustering, and its location information is obtained to initiate further attacks.
- We conducted a performance evaluation on TFRA under three typical ferry node path planning schemes, the results show that TFRA can identify ferry nodes with high accuracy.

The rest of this paper is organized as follows. Section 2 outlines some related works on ferry path planning and location anonymity. Section 3 describes prereq-

uisites including the network model, adversary model and ferry path planning schemes we used. Section 4 describes the details of TFRA. Section 5 analyzes and discusses our experimental results. Finally, Sect. 6 concludes this work.

2 Related Works

This paper is concerned with anonymity of nodes' location or indistinguishability of trajectory, which is similar to location privacy attacks on source nodes and sink nodes in wireless sensor networks, this type of attack requires finding the exact location of the source node or sink node. As far as we know, there are few studies on the attack of identifying ferry nodes in UAV network. Because the movement patterns of ferry and ordinary nodes are different, we aim to distinguish them from the perspective of trajectory, therefore we studied ferry path planning schemes as well.

2.1 Anonymity of Nodes' Location

To obtain the exact location of key nodes, the researchers suggested many types of attacks, such as passive attacks like wireless eavesdropping attacks, retrospective attacks, packet analysis attacks, and traffic analysis attacks; on the other hand, active attackers capture or cloning legal nodes to obtain the position information about a source or a base station [16].

In addition, according to the range of an adversary's monitoring area, the attack model can be classified into two categories: global attack and local attack. Local attackers in [13, 24] can only monitor a few hops and track the angle and strength of the received signal with antennas and spectrum analyzers. Global attackers in [14, 25] attack by monitoring the traffic of the entire network, analyze the content correlation, time correlation, and packet sending rate of the data packets.

2.2 Message Ferry Path Planning Schemes

The path planning schemes of ferry nodes can be divided into two categories: random path and planned path. In the case of the random path, the ferry node moves randomly and collects data from nodes within its communication range, for instance, MULEs in [23].

The planned path can be roughly divided into three categories: fixed path planning, global path planning, and distributed dynamic path planning.

Fixed Path Planning: the paths of all nodes are preset during network initialization to complete data collection and transmission, and the nodes move according to the preset paths during network operation [8, 27].

Global Path Planning: the basic idea is to determine a series of points that the path of ferry node must contain through factors such as distance, energy, and node movement model. *Bin Tariq M M et al.* [11] propose an algorithm

that ferry node passes a path which makes it touches each mobile node with a certain minimum probability. Similarly, in [26], the ferry nodes visit the exact rendezvous to collect data. In [5, 17, 22], the distance indicated by the hops, the number of packets forwarded to the nearest point by each node, energy, and total travel time of ferry node are suggested as factors to determine rendezvous.

Distributed Dynamic Path Planning: the above methods are all based on the prior knowledge (i.e. movement model) of other nodes, otherwise, they are not applicable. *Chin-Lin Hu et al.* in [15] designed a dynamic ferry node scheduling method that multiple ferry nodes tried to perceive the strength of information exchange in geographically adjacent places, found hot spots, and then independently planned travel paths through these hot spots. *JinYi Yoon et al.* [28] proposed a distributed path planning algorithm based on the weighted sum of travel time and the delivery deadline.

3 Network Model and Prerequisites

In this section, we will introduce the network model and prerequisites of message ferry path planning schemes.

3.1 Network Model

What we are considering is to apply UAV networks to search and rescue missions. As shown in Fig. 1(a), multiple ordinary searching nodes are sent to different locations in the mission area, the ferry nodes are introduced to establish connections between the ordinary searching nodes and the ground station. Data transmissions between nodes and the ground station use high-throughput links with limited range (Data Link). In addition, remote links (Control Link) are used for telemetry and control information.

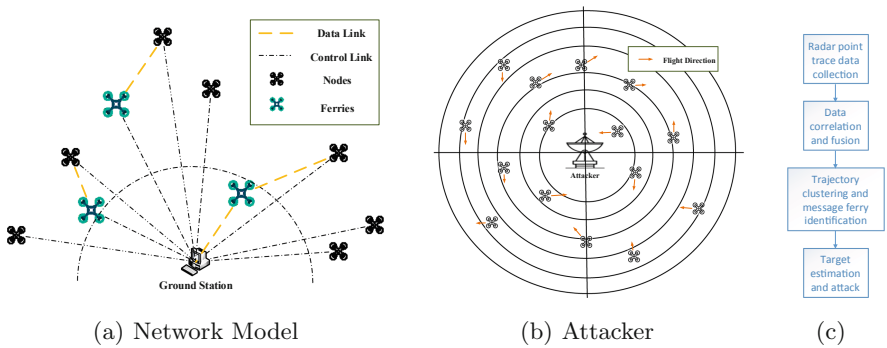


Fig. 1. Network and adversary model

Our network model can be abstracted into a weighted undirected graph G , which contains m mobile nodes, $G = \langle N, E \rangle$, $N = \{n_i | 1 \leq i \leq m\}$, $E = \{e_{ij} | 1 \leq$

$i, j \leq m\}$. e_{ij} indicates that nodes n_i and n_j are within the communication range of each other. Each node has a geographic location (latitude, longitude, height), direction, and speed, making it possible to predict future movements. It is assumed that message unicast is used and the ferry node only carries or forwards messages and doesn't generate any messages. The message transmission time is ignored because it is much shorter than the time for the UAV to move between nodes.

3.2 Message Forwarding with Ferries Path Planning

In this paper, we mainly consider three types of the planned path scheme.

Fixed Path Planning: we use the network scenario in [8,27], the number of UAVs in the network ranges from 13 to 24, the number of ferry nodes is 4, the position and trajectory of the UAVs are both preset in advance, the ordinary searching node is placed outside the communication range of the ground station, but within the communication range of at least one ferry node, the symmetric ferry node in the network is always moving towards the opposite direction to generate more paths for forwarding decisions.

Global Path Planning [5,17,22,26]: we summarize the common points of global path planning in several references: it is assumed that the ground station continuously receives and buffers the data transmission requests from the UAVs. When the data transmission requests accumulate to a certain number or reach a transmission cycle, the ground station plans the flight path of the ferry node. When there are multiple ferry nodes, the ground station first allocates the tasks, clusters the tasks according to geographic location, and then finds the closest ferry node to the clustering center, then the tasks in this cluster are assigned to that ferry node. When there is a ferry node closest to multiple clustering centers, multiple clustering results are merged and then assigned to that ferry node. Ferry node then plans a optimal path to complete tasks.

Distributed Dynamic Path Planning: we adopt the scheme in [15], and during initialization, all UAVs are randomly distributed in a fixed area.

3.3 Adversary Model

The attacker's goal is to disrupt or affect the communication in the UAV networks then ultimately paralyze the network. We assume that the attacker equipped with radar, radiofrequency, and sensors observes near the UAV networks. The maximum detection distance of radar is 3 to 10 km [10] and the attacker also uses radiofrequency and sensors to compensate for radar blind spots.

Such that the attacker can collect UAV point trace data and then obtain the flight trajectory by correlating the data. Then the attacker analyzes these flight trajectory, distinguishes the trajectory of the ferry node through the trajectory clustering algorithm, and accurately estimates the location of the target (ferry node), then launches further attacks [2]. The attacker model is shown in Fig. 1(b), the attack method and process are shown in Fig. 1(c).

4 TFRA: Trajectory-Based Message Ferry Recognition Attack

In this section, we describe in detail the steps of our proposed trajectory-based message ferry recognition attack. After we obtain the flight trajectories of UAVs, we aim to distinguish the trajectory of the ferry node by analyzing the trajectory data, so as to identify the ferry node.

4.1 Trajectory Representation and Preprocessing

We represent the trajectory by its spatial position at each time interval in the form of $[t_i, x_i, y_i, z_i]$, $i = 1, 2, 3, \dots$, where t_i represents the point-in-time and x_i, y_i, z_i represent coordinates of the position at time t_i .

To better represent the high-level meaningful movement patterns of the original trajectory, we convert the mission area to a pre-defined grid-based topology and transform the trajectory into a coarse-grained representation based on grid, that is, defined as a sequence of virtual grid *IDs* which are transformed from the corresponding coordinate points of the trajectory.

The paths of ferry nodes and ordinary nodes have different characteristics, including shapes of the trajectories curve and the flight area covered by the trajectories. Therefore, we normalize the trajectories by unifying the start point of each trajectory to remove the influence of the coordinates.

The clustering of the trajectories is more focused on the difference in movement patterns between trajectories. Therefore we pruned the trajectory to eliminate the impact of time. If nodes are in the same grid in continuous-time, only one grid is retained. The trajectory after pruning pays more attention to spatial distribution, and the removal of some points can also improve the clustering speed.

4.2 Trajectory Similarity Calculation

Trajectory similarity needs to be calculated to facilitate trajectory clustering, we adopted three typical similarity metrics including Euclidean distance [12], modified Hausdorff distance and Frechet distance [6], and the performance of these three metrics are evaluated through experiments in Sect. 5.

Hausdorff distance is generally used to measure the maximum mismatch between two trajectories, but it is defined on an unordered set of points, so even two completely different trajectory distances are likely to be very close. Therefore we adopt modified Hausdorff distance that considers the time series of the trajectories. As described in [9], for any point a_i in trajectory set A , first find the corresponding point b_i in trajectory set B . The search for corresponding point b_i is based on the length of the trajectory, according to the proportion of the distance from the starting point to the point a_i of the trajectory A to the total length of the trajectory A , find the corresponding point b_i in trajectory B that has the closest proportion. Then preset a range to find the neighbor

node set N_{b_i} of b_i , then the distance between point a_i and set B is $d_i = \min \{d(a_i, b_j) \mid b_j \in N_{b_i}\}$. The distance between two points is calculated using the euclidean distance.

For two trajectories $t_1 = \{grid_i, grid_{i+1}, \dots\}$ and $t_2 = \{grid_j, grid_{j+1}, \dots\}$, we use the coordinates of the grid center to represent the grid, the distance between two grids is calculated according to the coordinate of the grid center.

4.3 Clustering and Identification

Our purpose is to distinguish ferry nodes and ordinary nodes, so first we need to distinguish the trajectories of ferry nodes and ordinary nodes. Trajectory clustering is used to cluster similar trajectories into one class. Some common clustering methods are adopted as follows: agglomerative clustering, GMM and Kmeans, and their performance will be evaluated through experiments.

After the trajectory clustering is completed, we need to distinguish the clusters to find which type is the trajectory set of the ferry node. In order to distinguish the trajectories of the two types of nodes, we use the ferry path planning scheme mentioned in Sect. 3.2 to generate a series of trajectory data for ordinary nodes and ferry nodes. We analyze these trajectories and extract the following trajectories features: (1) trajectory length, (2) the proportion of the minimum/maximum coordinates of the trajectory to the maximum/minimum coordinates of the flight region, (3) the proportion of the number of grids covered by the trajectory to the total number of grids, (4) the starting point of the trajectory, (5) the ending point of the trajectory, (6) the proportion of the maximum span of coordinates of the trajectory to the maximum span of the flight region, (7) the number of changes in the trajectory slope.

In the following we use machine learning methods to determine the importance of these features, the methods used are **SelectKBest** and **Recursive feature elimination(RFE)** [3], and the importance of each $feature_i$ is regarded as its corresponding weight w_i , the first k features selected based on weight are used as the features for subsequent judgment clustering. Then we extract the k features of each trajectory in the aforementioned cluster and take the average of the k features of all trajectories in the cluster as the feature of the cluster, we use KNN, SVM, Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), etc. to train the above-mentioned trajectory feature set, and take the features of the cluster as input and judge whether each cluster is ferry nodes set or ordinary nodes set. The accuracy of these algorithms will be compared through experiment and it is found that the model trained by the random forest algorithm has high accuracy rate, so we use the random forest algorithm in the follow-up.

5 Performance Evaluation

In this section, we analyze and discuss the results of our proposed trajectory-based message ferry recognition attack scheme under three ferry node path planning methods, respectively.

Table 1. Parameters of three path planning scheme

	Fix_1	Fix_2	Fix_3	<i>Global distributed dynamic</i>
Simulation area (m^2)	800×800	1100×1100	1200×1200	800×800
Number of ordinary UAVs	9	11	20	20
Number of ferry UAVs	4	4	4	1 to 4
UAV speed (m/s)	5	5	5	5
Grid size (m)	20			
Communication range (m)	200			

5.1 Experiment Scenarios and Parameters

As shown in Sect. 3.2, we conducted simulation experiments on three types of ferry node path planning methods. The detailed experimental parameters of the three schemes are summarized as follows.

Fixed path planning: we carried out experiments in three scenarios with fixed path planning in [8, 27], the specific parameters are shown in Table 1.

Global path planning and distributed dynamic path planning: the specific parameters are shown in Table 1. The ordinary nodes adopt random waypoint model or random range move model. In the random waypoint model, ordinary nodes can move randomly in the entire area. In the random range move model, ordinary nodes are only allowed to move randomly within a range. The ferry node moves according to the flight path control information sent by the ground station, otherwise, it adopts the random waypoint model or remains still. For different network settings, 20 networks are randomly generated, and the final result is averaged.

Experimental variables: we study the impact of four types of experimental variables, including type of trajectory, clustering algorithm, ordinary node movement mode, and similarity metric. The specific variables are shown in Table 2.

5.2 Performance Metrics

We use the following indicators to evaluate performance:

- **Accuracy:** accuracy is defined as the probability of correctly identifying the ferry node. This indicator can evaluate the feasibility of the trajectory-based message ferry identification attack method.
- **Recognition Rate (RR):** the recognition rate is defined as the proportion of the identified ferry nodes to all ferry nodes. The greater the number of ferry nodes that are attackable and the higher the chance of attackers making the network paralyzed.

Table 2. Experimental variables

Experimental variables	Parameters
Type of trajectory	1. Original trajectories 2. Coarse-grained and normalized trajectories 3. Fully processed trajectories (pruned trajectories)
Clustering algorithm	1. Kmeans 2. GMM 3. Agglomerative clustering
Ordinary node move mode	1. Random waypoint model 2. Random range move model
Similarity metric	1. Euclidean distance 2. Modified Hausdorff distance 3. Frechet distance

5.3 Accuracy Comparison of Different Models

In this part, we compare the accuracy of five classification models: KNN, SVM, Decision Tree (DT), Random Forest (RF), Logistic Regression (LR). We used the three ferry path planning algorithms mentioned above to generate 7200 trajectory data and trained them with the five classification algorithms mentioned above. The experimental results show that most of the algorithms can achieve high accuracy. Due to the length of the paper, we only give the experimental results of random forest in the subsequent experiments.

5.4 Simulation Experiment Results and Discussion

The Impact of Trajectory Clustering Algorithm and Trajectory Pre-processing: the trajectory data set used by the trajectory clustering algorithm is divided into three types, *Original* represents the original trajectory, *Normalization* represents the coarse-grained and normalized trajectory, and *Pruning* represents the fully processed trajectory. The similarity metric is euclidean distance. As shown in Fig. 2(a) that coarse-grained representation and normalization can significantly improve the accuracy of recognition. Pruning does not affect the accuracy of recognition, and this operation can reduce the time of clustering.

Different clustering algorithms have little effect on the accuracy of recognition, in order to simplify, in the subsequent comparison experiments, *Agglomerative Clustering* is adopted.

The Impact of Ferry Path Planning Scheme: we obtain the trajectories of UAVs under three different path planning algorithms: fixed path planning (*Fix*), global path planning (*Global*), and distributed dynamic path planning (*Distributed*). As can be seen from the results in Fig. 2(b), different ferry node path planning algorithms have no obvious effect on the results of cluster recognition, accuracy is high in all three scenarios.

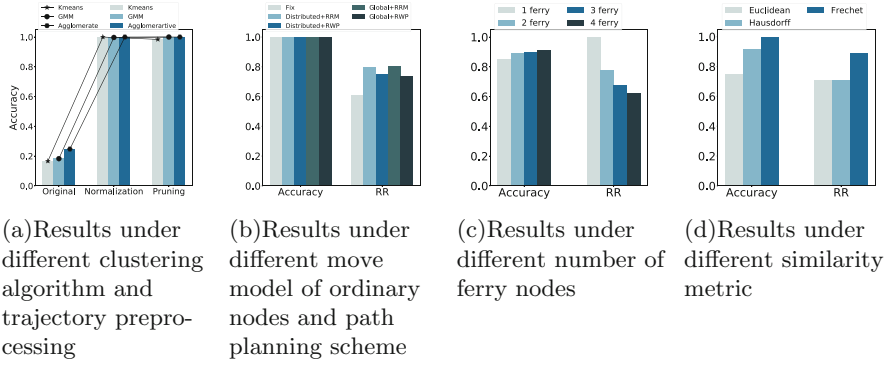


Fig. 2. Experiment results

The Impact of Movement Mode of Ordinary Nodes: in this part, we study the impact of two typical movement modes of ordinary nodes: random waypoint model (*RWP*) and random range model (*RRM*). It can be seen from the results in Fig. 2(b) that the movement mode of ordinary nodes does not have much impact on recognition results, but when using *RRM*, the recognition rate is higher. This is because the moving range of ordinary nodes in *RRM* is small, and the trajectory similarity to the ferry node is lower.

In the above two experiments, three similarity metrics were adopted, the number of ferry nodes ranged from 1 to 4, and the trajectory data set was the trajectory after pretreatment.

The Impact of the Number of Ferrying Nodes: in the experiment, the number of ferry nodes is increased from 1 to 4. As can be seen from the results in Fig. 2(c) that as the number of ferry nodes increases, the accuracy of recognition continues to increase, while the recognition rate decreases. As the number of ferry nodes increases, the number of identifiable ferry nodes increases, and the similarity of elements in ferry clustering also increases, so the recognition accuracy increases, and at the same time, the difficulty of identifying all ferry nodes increases, so the recognition rate decreases.

The Impact of Similarity Metric: Euclidean distance, Frechet distance, and modified Hausdorff distance are respectively expressed by *Euclidean*, *Frechet* and *Hausdorff*. From the results in Fig. 2(d), it can be seen that the accuracy is much higher than the Euclidean distance when the modified Hausdorff distance and Frechet distance are used as the similarity metric, which can be basically close to 1. At the same time, using the modified Hausdorff distance and Frechet distance as the similarity metric, the recognition rate is also greatly improved, the highest average recognition rate can reach 0.88. Among the three methods adopted, the Euclidean distance only considers the actual distance between the coordinate points; the modified Hausdorff distance considers the maximum mismatch between the two trajectories considering the time factor; the Frechet distance gradually analyze from global to local which avoids judging the similar-

ity of each sub-target only from the distance of point data, the closer the overall situation, the higher the similarity. Therefore, it can be seen from the experimental results that the Frechet distance has the best accuracy, followed by the Hausdorff distance, at the same time it has the reverse order in computing time. The descending order of recognition rate is *Frechet* > *Hausdorff* > *Euclidean*, but in terms of clustering time: *Frechet* > *Hausdorff* > *Euclidean*.

In the above two experiments, three similarity metrics were adopted, the number of ferry nodes ranged from 1 to 4, and the trajectory data set used by the trajectory clustering algorithm is divided into three types, including the same three types as in experiment one.

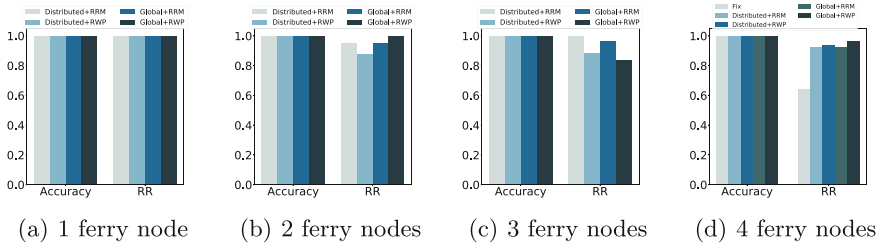


Fig. 3. Results on Agglomerative clustering with Frechet distance

Combining all the above experiments, we can know that the factors that have the greatest impact on recognition results are similarity metric and the number of ferry nodes, where the number of ferry nodes mainly affects recognition rate, and similarity metric has a great impact on both accuracy and recognition rate. Among all the experimental results, the results using *Agglomerate clustering* and *Frechet* distance are the best. The specific results of the experiment are shown in Fig. 3. The recognition rate of global path planning methods and distributed dynamic path planning method is above 90%, the average recognition rate reaches about 95%, and the accuracy is close to 1. For the fixed path planning, the accuracy is close to 1, and the recognition rate is above 60%.

6 Conclusion and Future Work

Ferry nodes play an important role and the exposure of their locations may cause serious consequences. Therefore in this paper, we propose a Trajectory-based message Ferry Recognition Attack (TFRA), using the difference between the flight mode of the ferry node and the ordinary node to identify ferry nodes in UAV network and get its location information to launch further attacks. At the same time, we have summarized ferry path planning schemes and performed performance evaluation of TFRA on three typical ferry path planning schemes, the results show the feasibility of our proposed method. In the future, we will investigate the online identification of ferry nodes, improving accuracy and recognition rate is also our goal.

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